KLAi

Inference: is the difference real, or could it be luck?

When two groups differ, how do I know the gap is real and that chance does not explain it?

AI and Data Science Professional | Session 12 of 24 | Module C: Analyse and explain
KLAi Professional Learning | Jotham Martin Wambi

Today in one view: what you will leave with

By the end of today you can



Compute and read a standard error, a 95 percent confidence interval and a bootstrap interval



Run and read a permutation test, a Welch two-sample t-test and a test of two proportions, with the null hypothesis and the p-value



Report a decision with its effect size, and avoid false positives from many tests and from stopping early

Your two hours: concepts first, then a Python lab in Colab

0:00	10	Recall and the question of the day
0:10	45	Concepts: picture, plain words, one example by hand
0:55	10	Break
1:05	45	Lab: predict, run, read, change
1:50	10	Exit check, keep and next

Ideas and pictures first. Then you open the notebook and test the offer yourself.

Three questions from last time

1. Branch B01 has the most returns. Is it the branch with the worst returns problem?

2. What three parts does every finding need?

3. Which chart shows whether two numeric variables move together?



Answer aloud first. Then we check together.

Answers to the three questions from Session 11

	Question	Short answer
	**Branch B01 has the most returns. Is it the branch with the worst returns problem?**	Not yet known. Divide by lines sold to get the rate: B01 has 1.67 per 100 lines and B07 has 3.18, the highest.
	**What three parts does every finding need?**	A number, a comparison and a limit.
	**Which chart shows whether two numeric variables move together?**	A scatter plot, drawn with sns.scatterplot. The correlation r puts a number on it.

Words we use today

	Word	What it means
	**Standard error (SE)**	How much a sample mean varies from one random sample to the next. It is the standard deviation divided by the square root of the sample size.
	**Confidence interval (CI)**	A range built from the data. A 95 percent interval is the estimate plus or minus 1.96 standard errors.
	**Null hypothesis (H0) and p-value**	H0 says there is no real difference. The p-value says how often chance alone would give a gap at least this big if H0 were true.
	**Effect size**	How big the gap is, in units a person can feel, such as shillings or percent. Practical significance asks whether that size matters.
	**False positive (Type I error)**	Declaring a difference when there is none. Running many tests, or stopping early, makes it far more likely.

An A/B test: chance decides who gets the offer (treatment) and who does not (control)

	A	B	C	D
	customer_id	group	next_month_spend_ugx	returned_within_30d
1	C0001	offer	75,600	1
2	C0002	offer	70,100	0
3	C0003	offer	34,400	0
4	C0004	control	40,600	1
5	C0005	offer	26,600	1
6	C0006	offer	48,200	0

Random assignment makes the groups alike on average in everything except the offer.

Each record is one loyalty customer.

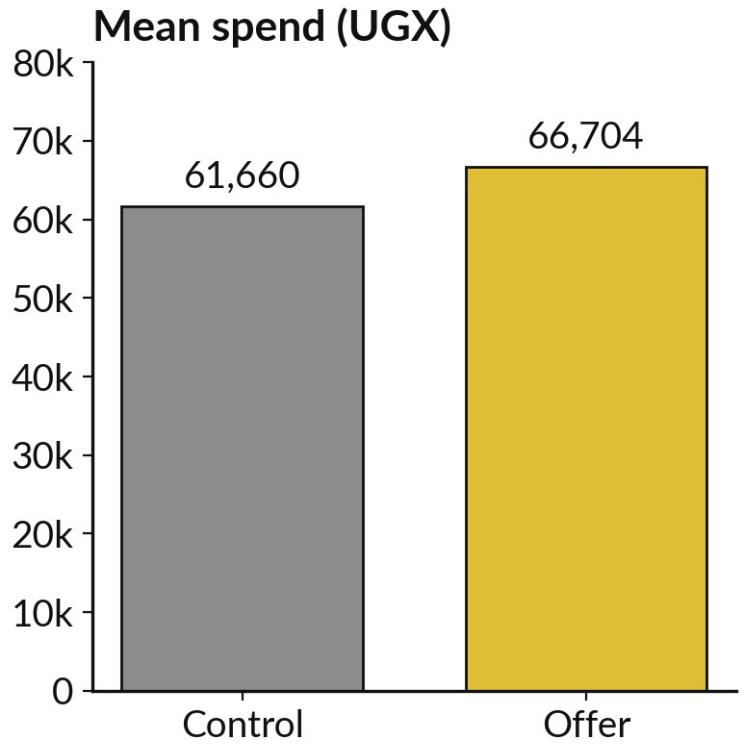
1,245 got the offer (treatment). 1,255 did not (control).

The outcome is next month spend.

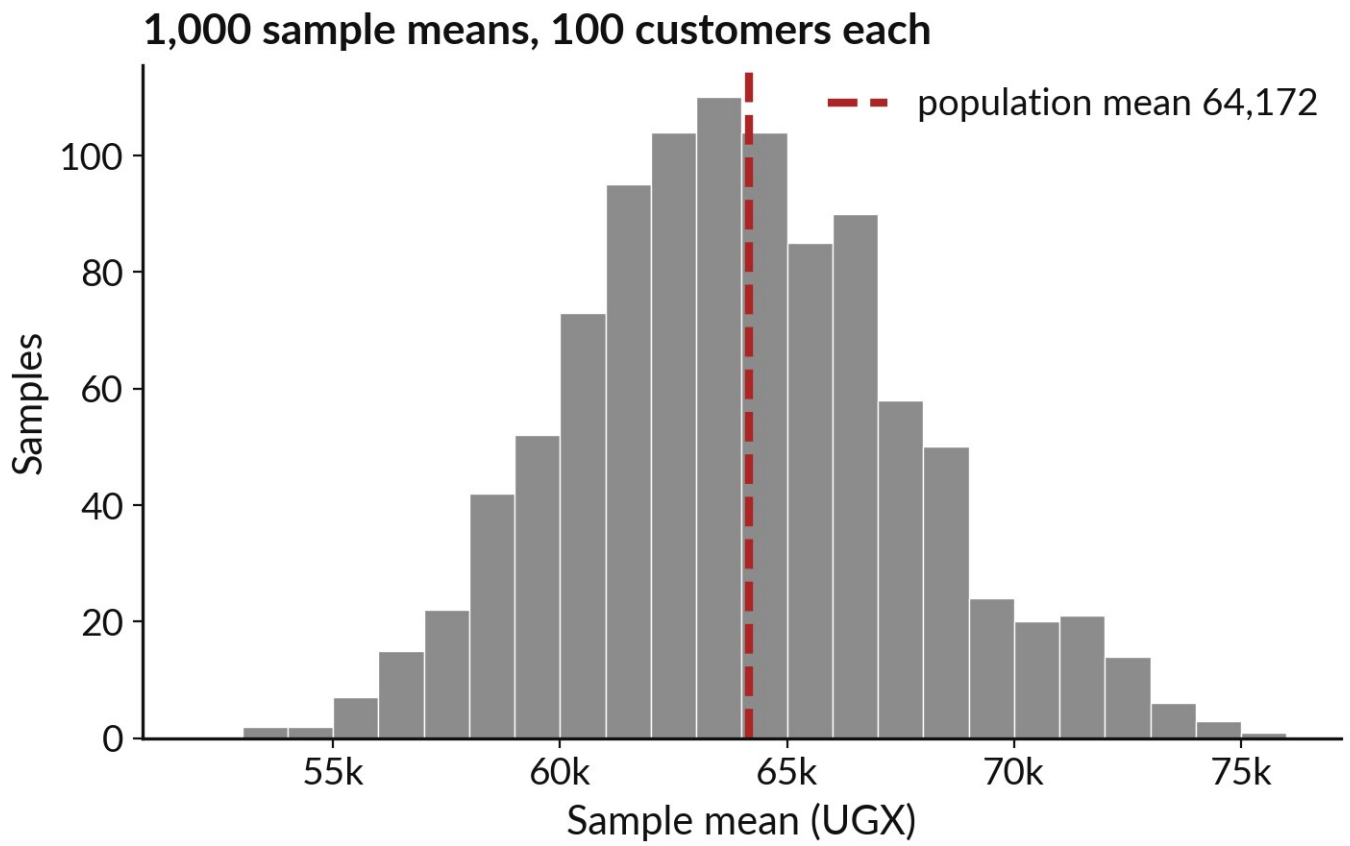
The lift in average spend is 5,044 UGX. Is it the offer, or luck?

5,044

UGX more per customer in the treatment group: a mean of 66,704 against 61,660 in the control group. The lift is the treatment mean minus the control mean.



Sampling variability: every random sample gives a slightly different mean



A sample mean is an estimate of the population mean. Its spread is the standard error.

Population: all 2,500 customers. Sample: 100 drawn at random, with replacement.

The 1,000 means range from 53,306 to 75,316.

Their standard deviation, 3,736, is the standard error.

The standard error of a mean is the standard deviation divided by the square root of n

Formula

$$\mathrm{SE} = \frac{s}{\sqrt{n}}$$

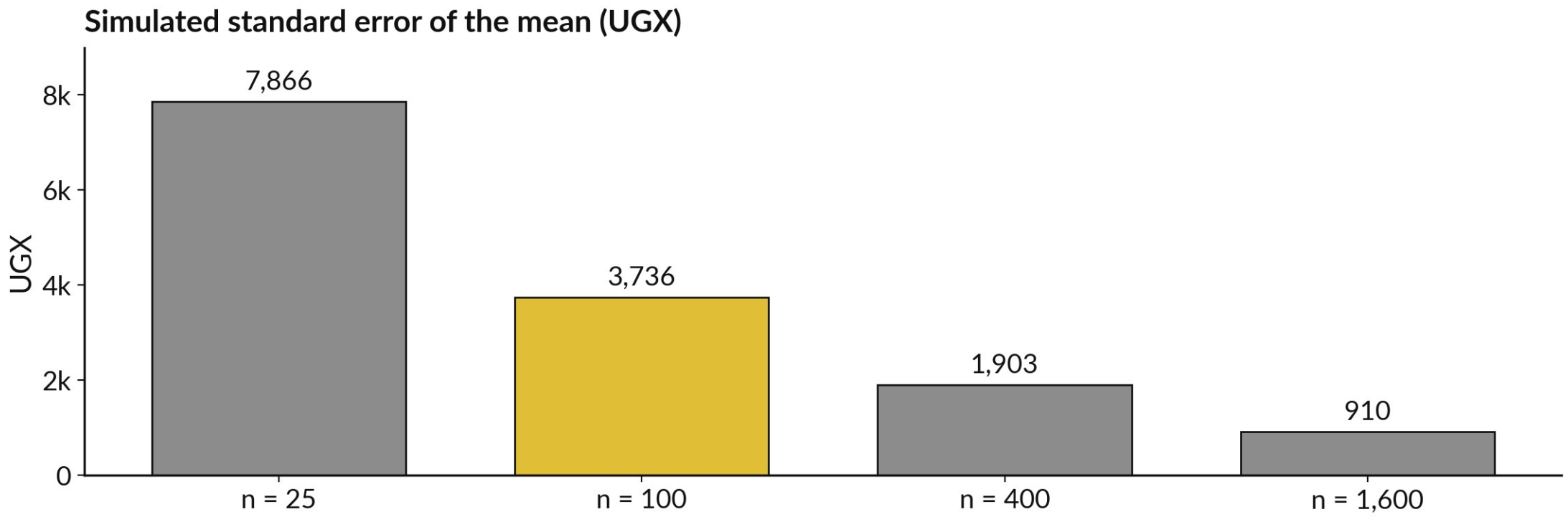
Control group

$$\frac{36,568}{\sqrt{1255}} = 1,032$$

Symbol	What it means	In our data
SE	The standard error: the typical distance of a sample mean from the population mean.	1,032 UGX
s	The sample standard deviation (ddof=1): the spread of customers.	36,568 UGX
n	The sample size: how many customers.	1,255 control customers

Excel: =STDEV.S(range)/SQRT(COUNT(range)). NumPy: x.std(ddof=1) / np.sqrt(len(x)).

A bigger sample has a smaller standard error: four times the customers halves it



Formula values: 7,640, 3,820, 1,910, 955. Each bar is 1,000 samples with replacement.

A 95 percent confidence interval is the estimate plus or minus 1.96 standard errors

Formula

$$\text{CI} = \text{estimate} \pm 1.96 \times \text{SE}$$

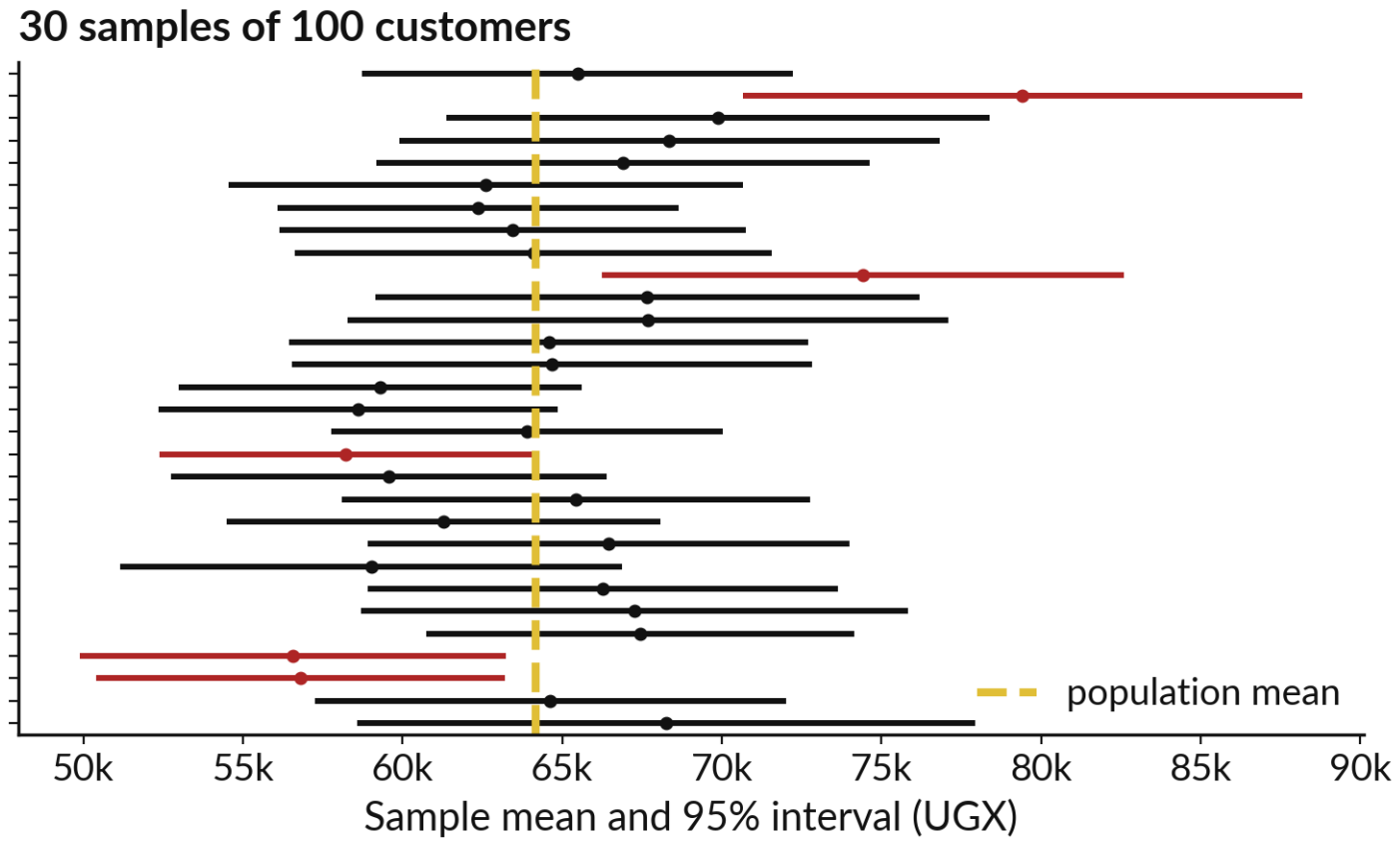
The lift

$$5{,}044 \pm 1.96 \times 1{,}526 = 2{,}054 \text{ to } 8{,}035$$

Symbol	What it means	In our data
estimate	The number from the sample. Here the lift.	5,044 UGX
SE	The standard error of the lift: the square root of the two squared standard errors added.	1,526 UGX
1.96	The critical value for a 95 percent confidence level, from the normal distribution.	95 percent
CI	The interval. It excludes zero when the lift is clearly above or below zero.	2,054 to 8,035

Each group alone: control CI 59,636 to 63,683. Offer CI 64,502 to 68,906.

What 95 percent means: it describes the method over many samples, not one interval



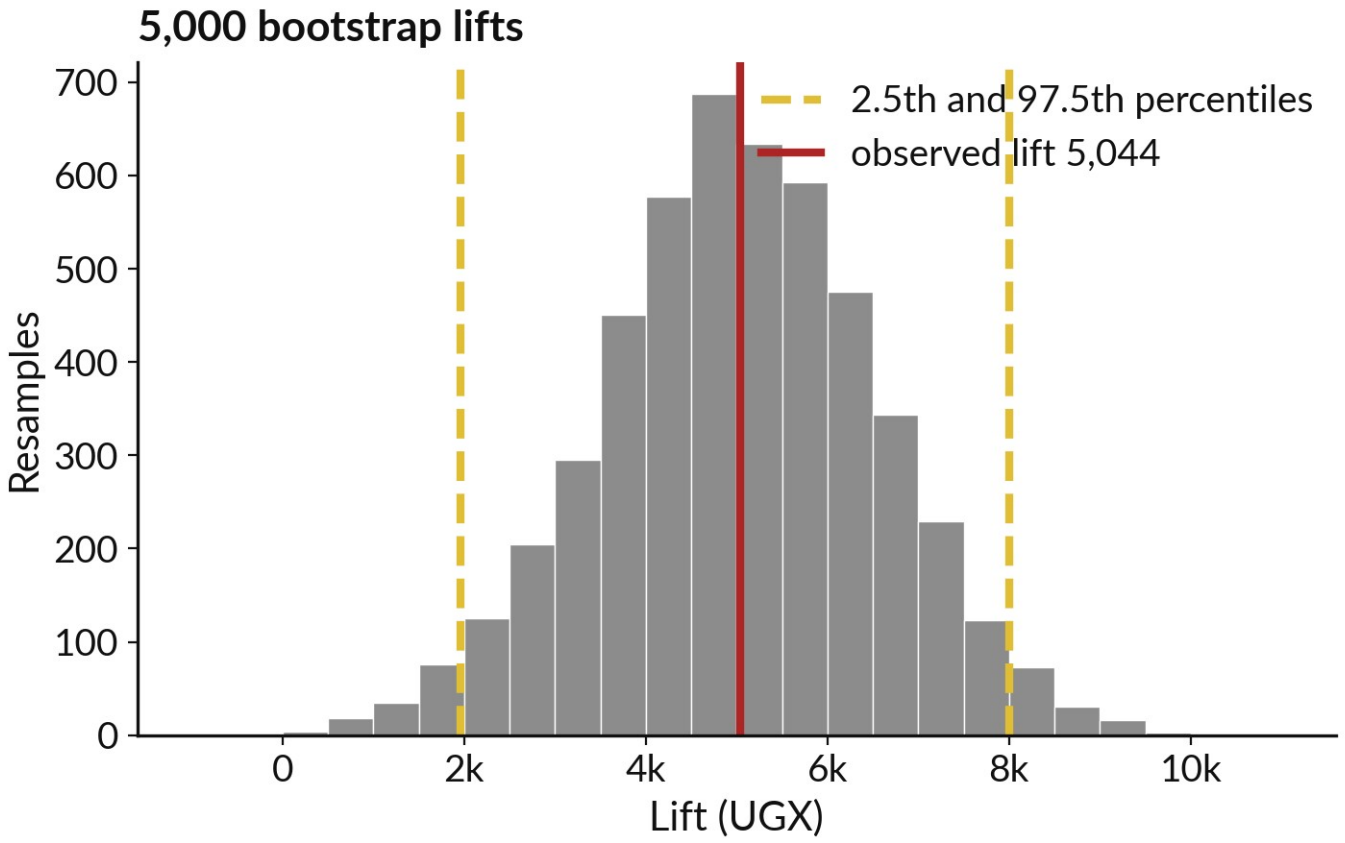
Draw many samples. About 95 in 100 intervals catch the population mean.

25 of these 30 intervals catch it. The red ones miss.

Over 1,000 samples, 93.9 percent did.

Spend is skewed, so small samples fall a little short of 95.

The bootstrap resamples your own data to measure sampling variability without a formula



Bootstrap: resample each group with replacement, same size, and recompute the lift. Repeat 5,000 times.

The spread of the 5,000 lifts is 1,522, close to the formula SE 1,526.

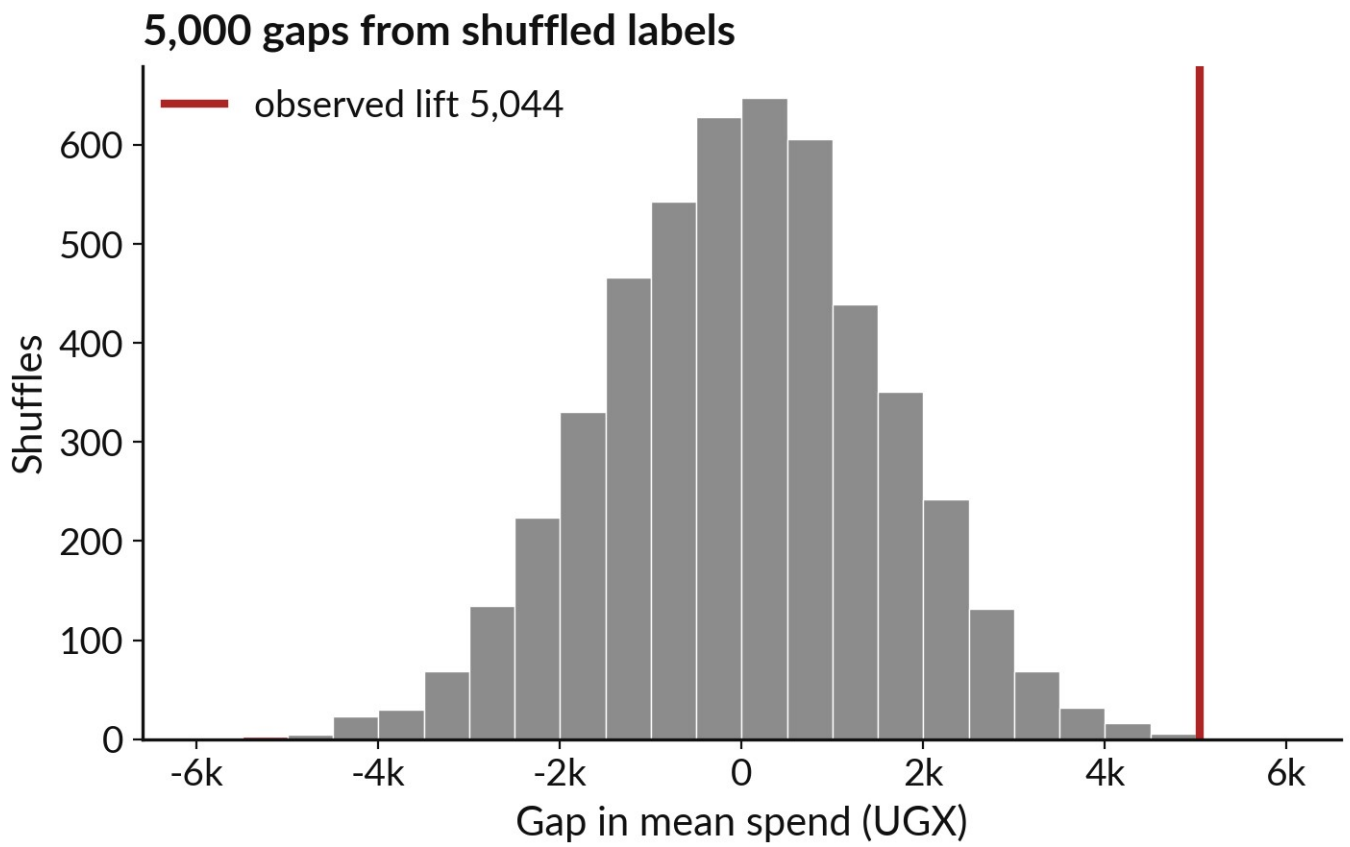
Percentile interval: 1,959 to 8,004.

It needs no formula, so it works for medians too.

A hypothesis test asks: how surprising are the data if nothing is going on?

Term	Plain meaning	In the offer test
Null hypothesis (H0)	There is no real difference. Any gap is chance.	The offer does not change average spend.
Alternative hypothesis (H1)	There is a real difference.	The offer changes average spend.
Test statistic	One number that measures the gap against the noise.	The lift, or t = 3.31
p-value	How often chance alone gives a gap at least this big, if H0 is true.	About 0.00096 (see the next slides)
Significance level (alpha)	The threshold chosen before the test. Usually 0.05.	alpha = 0.05
Decision	p below alpha: reject H0. Otherwise, keep H0.	0.00096 is below 0.05: reject H0

A permutation test shuffles the group labels to see what chance alone produces



Shuffle offer and control labels at random. If H0 is true, the labels carry no information.

Only 7 of 5,000 shuffled gaps were as big as 5,044 (red).

p = 7 / 5,000 = 0.0014, or 8 / 5,001 = 0.0016.

The p-value is the share of shuffles at least as extreme.

A p-value is easy to misread: know what it says and what it does not

It says

If H0 were true, data this extreme would be rare: p = 0.00096.

It does not say

The probability that H0 is true.

How big the effect is.

That the result will repeat.

A small p-value is evidence against H0. Effect size tells you how much it matters.

Welch's two-sample t-test: the lift divided by its standard error

Test statistic

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

Offer against control

$$t = \frac{5,044}{1,526} = 3.31$$

Symbol	What it means	In our data
x-bar 1, x-bar 2	The two group means.	66,704 and 61,660
s, n	Standard deviation and size of each group. Welch lets the two spreads differ.	39,649, 1,245 and 36,568, 1,255
df and p-value	Welch-Satterthwaite degrees of freedom, and the two-sided p-value from the t distribution.	2478.5 and 0.00096

SciPy: stats.ttest_ind(offer, control, equal_var=False) gives the same t and p-value.

A test of two proportions asks the same question for a yes or no outcome

Test statistic

$$z = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}\,(1-\hat{p})\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$$

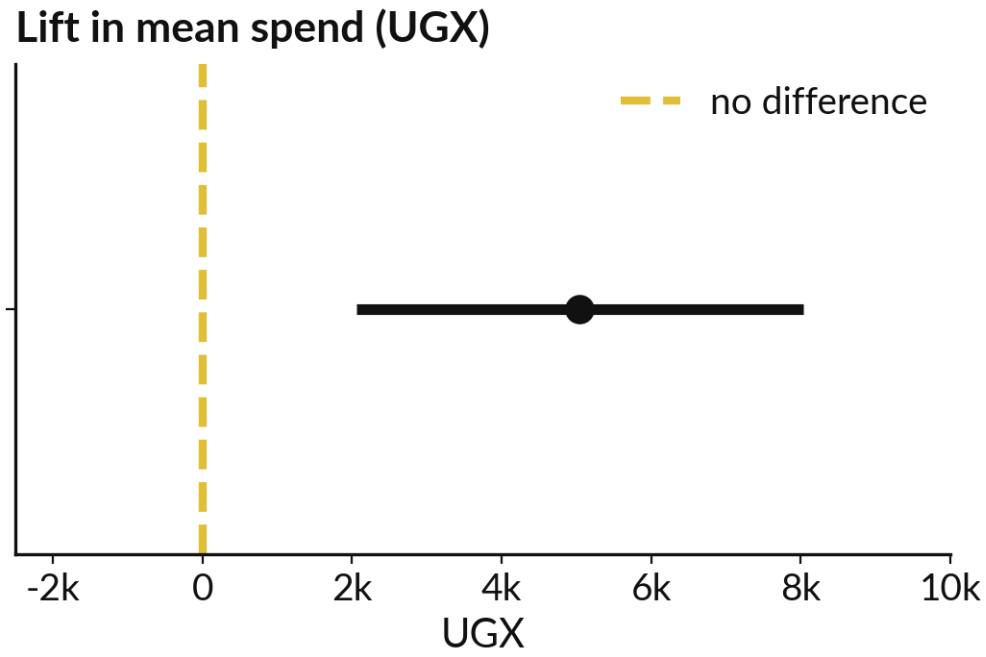
Came back within 30 days

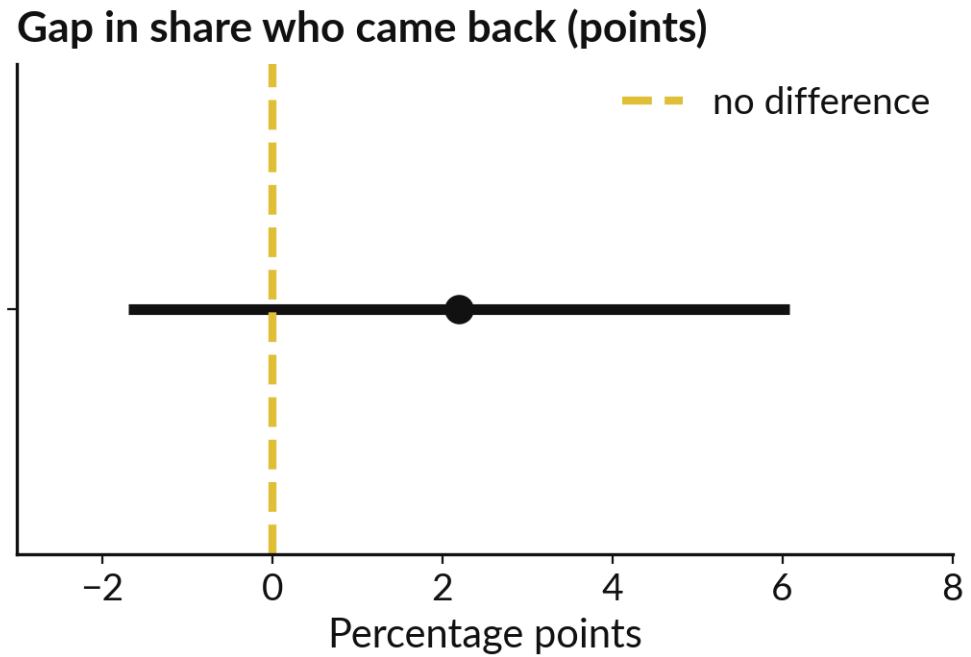
$$z = \frac{0.0219}{0.0199} = 1.10$$

Symbol	What it means	In our data
p-hat 1, p-hat 2	The share who returned within 30 days in each group.	0.4514 and 0.4295
p-hat	The pooled share: all returners over all customers. It is the rate if H0 is true.	0.4404
z and p-value	The test statistic, and the two-sided p-value from the normal distribution.	1.10 and 0.27

statsmodels proportions_ztest gives the same z. This z uses the pooled share. The interval after it does not.

The spend interval stays above zero.
The return-rate interval crosses zero.





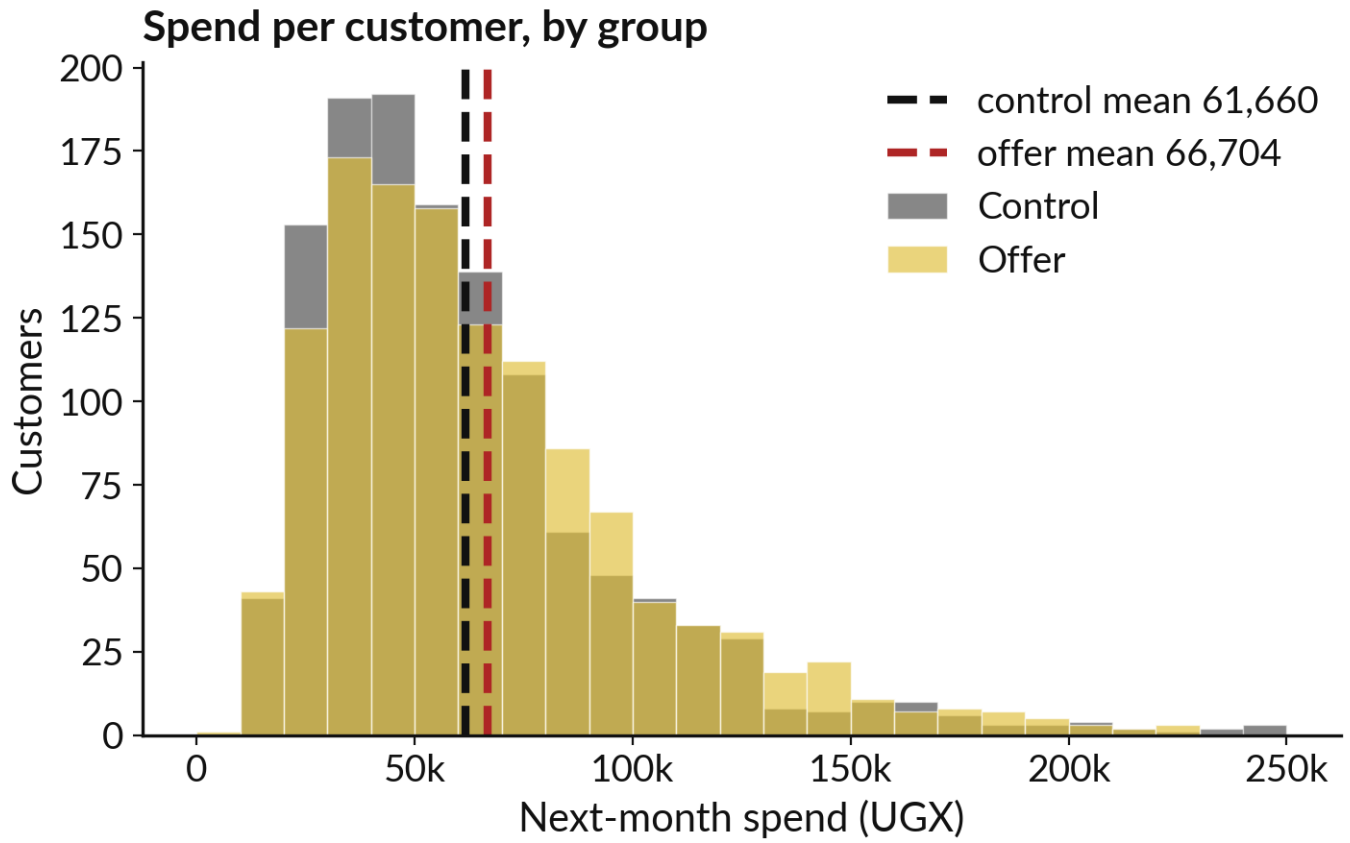
Spend: 2,054 to 8,035. Return rate: -1.7 to 6.1 points. 95 percent intervals, formula method.

Two questions, two different answers

	Question	Estimate and 95% interval	p-value	Decision
	**Did the offer raise next-month spend?**	+5,044 UGX (2,054 to 8,035)	0.00096	Reject H0. Strong evidence of a lift.
	**Did more customers come back within 30 days?**	+2.2 points (-1.7 to 6.1)	0.27	Keep H0. No evidence of a change.

Not rejecting H0 is different from proving the offer has no effect on returning. The interval is wide, and more customers could narrow it.

Effect size: the lift is small next to how much customers differ



The groups overlap heavily. Cohen's d = 0.13, below the usual 0.2 mark for a small effect.

Lift: 5,044 UGX, 8.2 percent of the control mean.

Cohen's d is the lift over the pooled standard deviation, 38,133.

Statistically clear, practically modest.

Statistical significance and practical significance answer different questions

Statistical significance

Is the gap bigger than chance? p = 0.00096, below alpha 0.05.

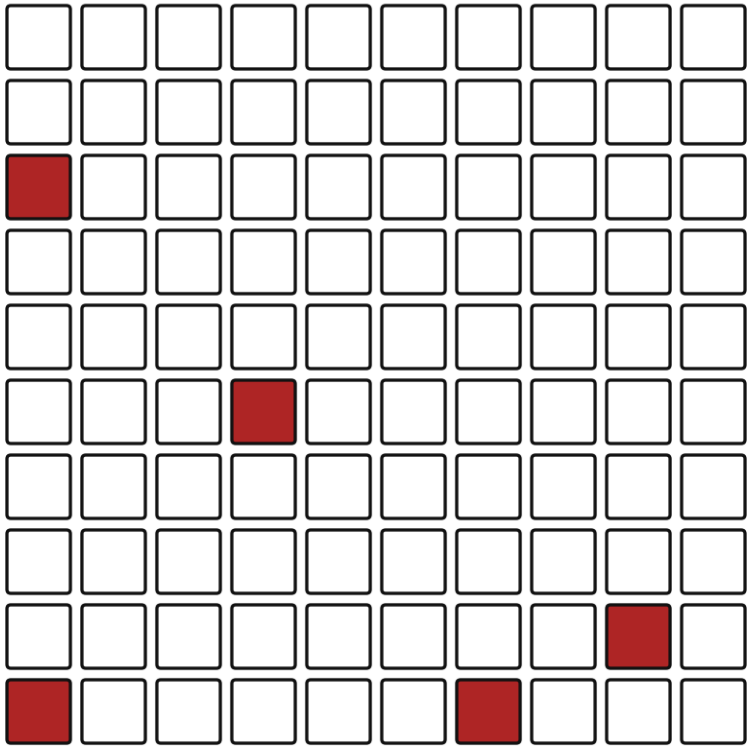
Practical significance

Hypothetical: cost 3,000 UGX per customer, margin 20 percent of extra spend.

20 percent of 5,044 is 1,009, minus 3,000: a loss of 1,991.

Report both: is it real, and is it big enough to act on?

A false positive (Type I error): a test can raise an alarm when nothing is going on



5 of 100 fake tests raise an alarm (red)

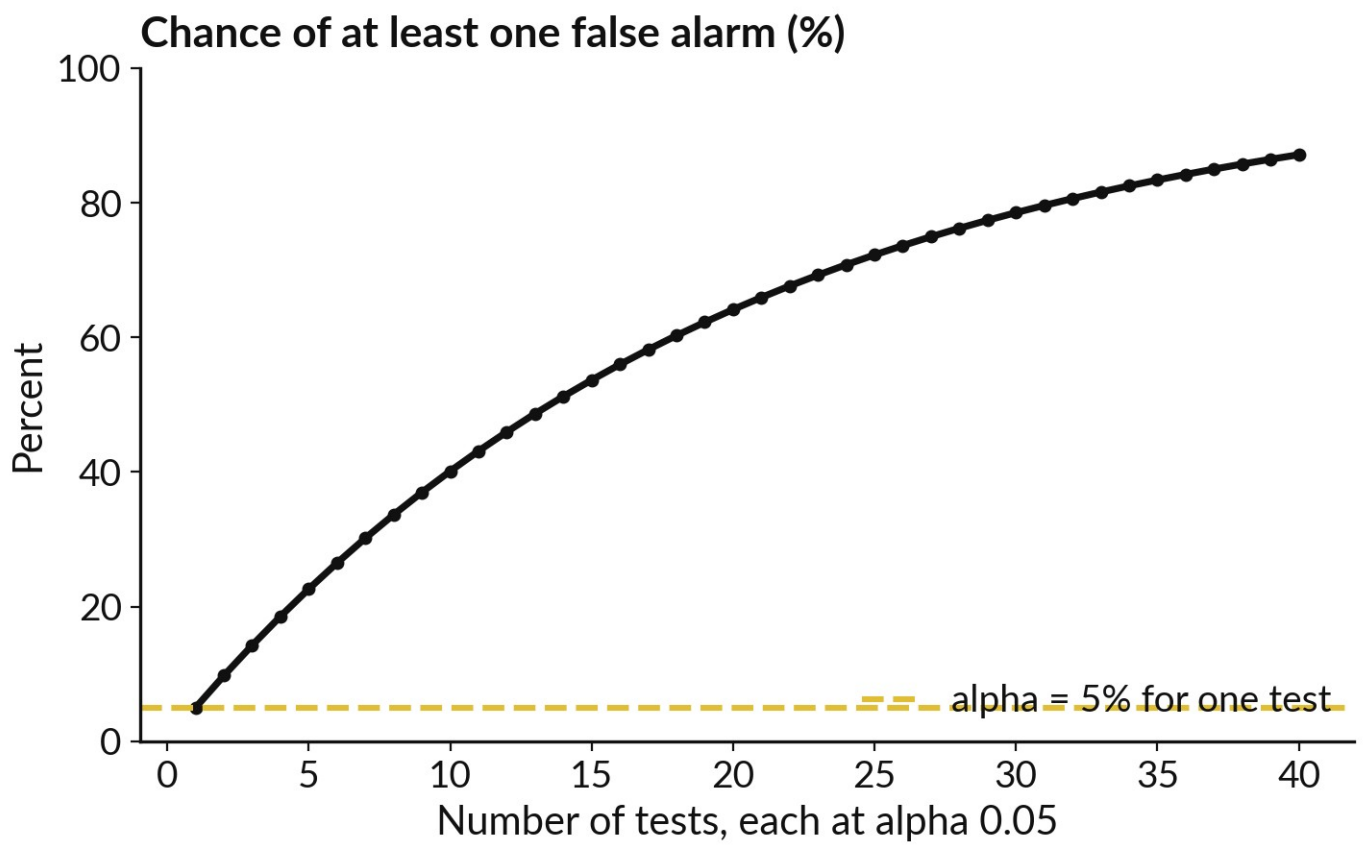
Alpha = 0.05 means about 5 in 100 tests of a true H0 give p below 0.05.

Fake test: split the control group at random into two halves. The true gap is zero.

5 of 100 splits had p below 0.05.

A false negative (Type II error) is a real effect that a test misses.

Multiple comparisons: run many tests and some alarms will be false



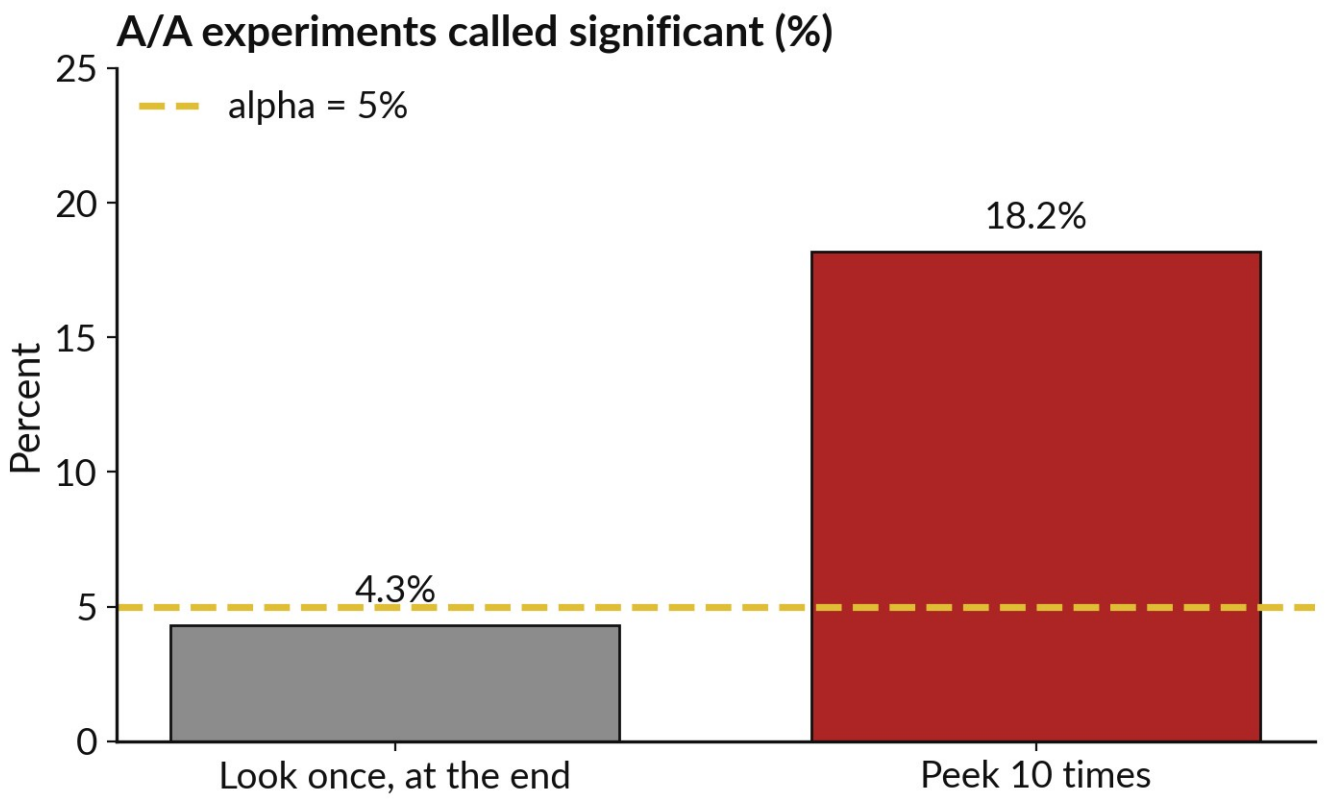
With m independent tests at alpha 0.05, the chance of at least one false alarm is 1 minus 0.95 to the power m.

5 tests: 23 percent. 20 tests: 64 percent.

Bonferroni correction: test each at alpha divided by m.

Decide the tests before you look.

Optional stopping: peeking at a running test and stopping when p is small inflates false alarms



Decide the sample size in advance. Stopping when p dips below 0.05 is a form of p-hacking.

2,000 A/A experiments, no real difference.

One look at the end: 4.3 percent significant.

Stop at the first of 10 looks with p below 0.05: 18.2 percent.

How many customers does the test need? About 914 per group to catch a lift of 5,000 UGX

	Customers per group	Margin of error of the lift (95% interval, plus or minus UGX)
	250	6,685
	500	4,727
	1,000	3,343
	1,245 (our offer group)	2,996
	2,500	2,114

Margin = 1.96 x s x square root of (2 / n), with s = 38,133 UGX. Power: with 914 per group, a test at alpha 0.05 catches a true lift of 5,000 about 80 times in 100 (power 80 percent, a common choice).

Where you will use this at work: check that a gap is real before you act on it

Your job	The comparison	What today lets you do
Marketing analyst at a telecom, using Python	A promotion sent to half the subscribers.	Report the lift with an interval and a p-value, and the cost per customer.
Credit risk analyst at a bank, using Python	Default rate in two branches or two scorecards.	Run a two-proportion test and report the interval together with the two rates.
Programme officer at an NGO	Test scores in a pilot school and a comparison school.	Use a permutation test when the groups are small or skewed.
Operations manager at a retailer	Return rates before and after a change in supplier.	Fix the test before the data arrive, and avoid peeking.

Read recipe 1 before the lab: the lift, its standard error, t and the interval by hand

```
import numpy as np
import pandas as pd
test = pd.read_csv('offer_test.csv')
spend = test.next_month_spend_ugx
offer = spend[test.group == 'offer'].to_numpy()
control = spend[test.group == 'control'].to_numpy()
lift = offer.mean() - control.mean()
v_o = offer.var(ddof=1) / len(offer)
v_c = control.var(ddof=1) / len(control)
se = np.sqrt(v_o + v_c)
print(round(lift), round(se), round(lift / se, 2))
print(f'{lift - 1.96*se:.0f} {lift + 1.96*se:.0f}')
```

```
5044 1526 3.31
2054 8035
```

In plain words

Split the spend into offer and control arrays.

v_o and v_c are the squared standard errors. Their sum under a square root is the standard error of the lift.

t is the lift over its standard error: 3.31.

The interval is the lift plus or minus 1.96 standard errors. SciPy ttest_ind with equal_var=False gives the same t.

Read recipe 2: the bootstrap interval, with a fixed seed

```
rng = np.random.default_rng(42)
boot = [rng.choice(offer, len(offer)).mean()
        - rng.choice(control, len(control)).mean()
        for _ in range(5000)]
low, high = np.percentile(boot, [2.5, 97.5])
print(round(low), round(high))
```

```
1959 8004
```

In plain words

default_rng(42) fixes the seed, so everyone gets the same random draws.

choice with a size draws with replacement. Each pass resamples both groups and takes the lift.

percentile cuts off the lowest and highest 2.5 percent.

The print is the 95 percent bootstrap interval: 1,959 to 8,004.

Read recipe 3: the permutation test shuffles the offer labels

```
values = spend.to_numpy()
is_offer = (test.group == 'offer').to_numpy()
rng = np.random.default_rng(42)
gaps = []
for _ in range(5000):
    s = rng.permutation(is_offer)
    gaps.append(values[s].mean() - values[~s].mean())
hits = (np.abs(gaps) >= abs(lift)).sum()
print(hits, hits / 5000)
```

```
7 0.0014
```

In plain words

values holds every customer spend. is_offer is True for the offer group.

rng.permutation shuffles the labels, keeping the group sizes.

Each shuffle gives one gap in mean spend. hits counts gaps at least as big as the real lift, ignoring the sign.

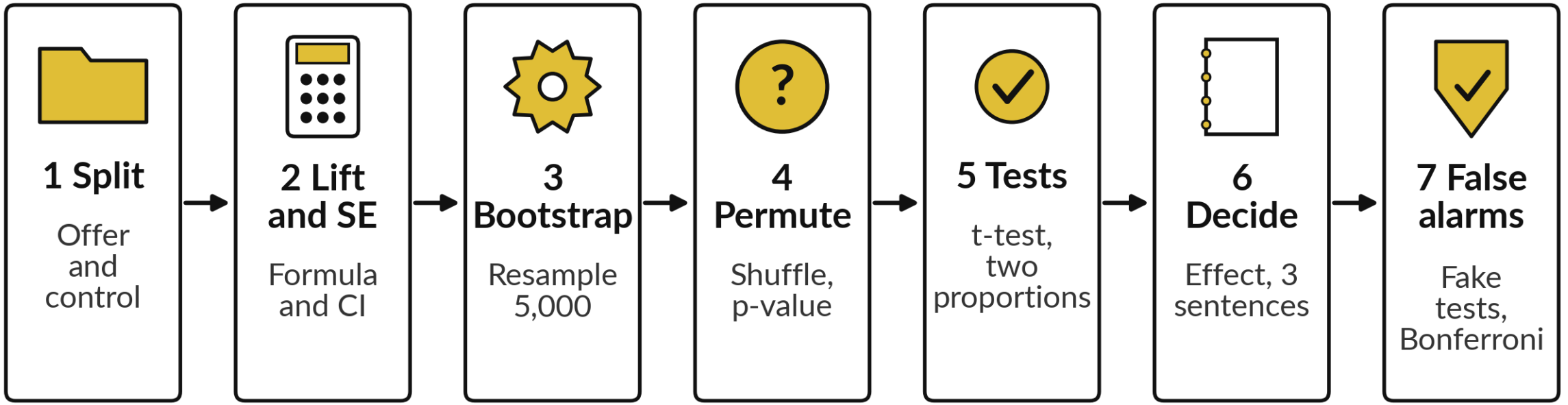
7 hits in 5,000 gives p = 0.0014.



Part 2: the Python lab

Close the slides. Open the notebook.

The lab has seven steps, and you keep a decision note about the offer



Twenty-one automatic checks in 45 minutes. Predict, run, read, change.

Check your numbers before you move on

What you compute	You should get
Customers in offer and in control	1,245 and 1,255
Mean spend in offer and in control (UGX)	66,704 and 61,660
The lift (UGX)	5,044
Standard error of the lift, and the 95% interval	1,526, and 2,054 to 8,035
Bootstrap interval (seed 42, 5,000 resamples)	1,959 to 8,004
Permutation test (seed 42): hits and p-value	7 of 5,000, p = 0.0014
Welch t statistic and p-value	3.31 and 0.00096
Two proportions: z and p-value	1.10 and 0.27

Bootstrap and permutation numbers repeat only with the same seed and the same order of calls.

Five likely mistakes in the lab and how to fix them

What you see	What it means	What to do
Bootstrap interval differs from the slide	A different seed or call order.	Make rng once, before the loop.
p-value is 0.0 from the permutation test	No shuffle was this extreme.	Report p below 1 in 5,000.
NameError: name 'stats' is not defined	The import cell was not run.	Run the first code cell: from scipy import stats.
t is 0.002 instead of 3.31	The lift was divided by the variance.	Take the square root: np.sqrt(v_o + v_c).
p-value is half of the expected value	A one-sided test was run.	Use the two-sided p-value: multiply the tail area by 2.

Exit check: three questions, then compare with the short answers

	Question	Short answer
	**A test gives p = 0.03 with alpha = 0.05. What do you conclude, and what do you still need to report?**	Reject H0: if H0 were true, a gap this big would be rare. Also report the effect size and the confidence interval.
	**You run 20 tests at alpha = 0.05 on a dataset with no real effects. About how many false positives do you expect?**	About one, because 20 times 0.05 is 1. The chance of at least one is about 64 percent.
	**A 95 percent confidence interval for the lift is -200 to 5,000 UGX. Is the lift significant at 0.05?**	No. The interval includes zero, so the data are compatible with no difference.

Save your notebook with its decision note

What you keep from today

Keep your inference notebook: the offer test done five ways (formula, bootstrap, permutation, t-test, two proportions) and a decision paragraph with the lift, its interval, the p-value and the limit.

Our question today

When two groups differ, how do I know the gap is real and that chance does not explain it?

Next session

Session 13: Regression. How much does one thing change when another changes? We model weekly orders and reach Project milestone 2.

Take away

Your lab file

Offer test notebook

Your writing

A decision note

Next session

Regression